

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 40%

Annexure A

Coding challenge

1. Develop a Python program that generates a Star Topology for a given number of hosts connected to a central switch. The program should accept the number of hosts as input and display the topology either in ASCII format or a graphical representation. For each host, print the host label, the name of the central switch, and the corresponding Cat6 UTP cable length connecting the host to the switch. Verify that all hosts are connected through the central switch and explain how the star topology facilitates reliable communication in a local area network.

CODE:

```
# Star Topology Generator
```

```
print("===== STAR TOPOLOGY =====")
```

```
# Get number of hosts
```

```
n = int(input("Enter the number of hosts: "))
```

```
switch = "Central-Switch"
```

```

cable_length = 10 # Cat6 UTP cable length in meters

print("\nSTAR TOPOLOGY CONNECTION DETAILS")

print("-----")

print("Host\t\tSwitch\t\tCable Length")

print("-----")

# Display connection details

for i in range(1, n + 1):

    host = "Host-" + str(i)

    print(host, "\t\t", switch, "\t\t", cable_length, "m")

# Verification

print("\nVerification:")

print("All", n, "hosts are connected to", switch + ".")

print("The star topology is successfully created.")

print("\nCommunication:")

print("Host → Central Switch → Other Host")

```

OUTPUT:

```

Output
===== STAR TOPOLOGY =====
Enter the number of hosts: 7

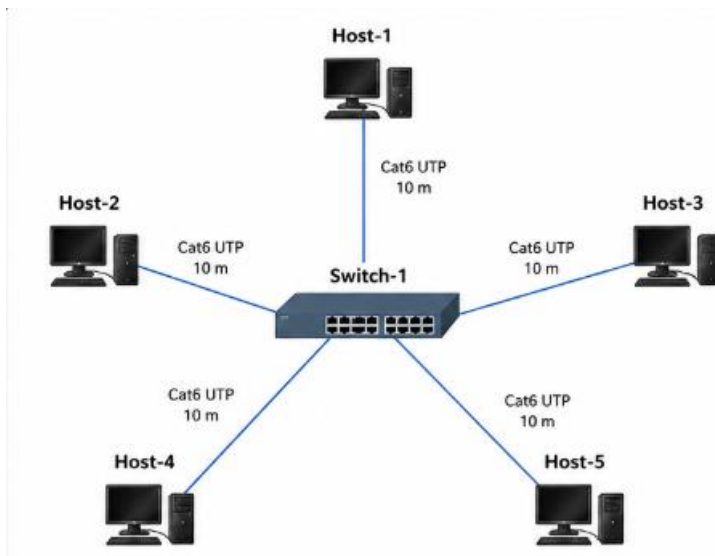
STAR TOPOLOGY CONNECTION DETAILS
-----
Host      Switch      Cable Length
-----
Host-1    Central-Switch  10 m
Host-2    Central-Switch  10 m
Host-3    Central-Switch  10 m
Host-4    Central-Switch  10 m
Host-5    Central-Switch  10 m
Host-6    Central-Switch  10 m
Host-7    Central-Switch  10 m

Verification:
All 7 hosts are connected to Central-Switch.
The star topology is successfully created.

Communication:
Host → Central Switch → Other Host

```

DIAGRAM



EXPLANATION:

Central Switch as the Node Hub

At the core of the topology sits the **Central-Switch**, acting as the central point of interconnection for all endpoints in the local area network (LAN). Every host (**Host-1** through **Host-7**) maintains a dedicated, direct point-to-point physical line connected exclusively to a specific port on the central switch.

Cat6 UTP Physical Cabling

Each connection utilizes Unshielded Twisted Pair (Cat6 UTP) Ethernet cables. These dedicated link runs carry point-to-point data signals isolated from other host connections, ensuring standard gigabit throughput without electrical signal collision risks over distances up to 100 meters.

Communication Workflow

- **Packet Generation:** When **Host-1** sends data to **Host-3**, the data frame travels directly over Host-1's dedicated Cat6 cable to the central switch port.
- **Intelligent Forwarding:** The central switch inspects the destination MAC address inside the Ethernet frame header and uses its internal MAC address table to forward the packet solely out of the port connected to **Host-3**.
- **Traffic Isolation:** **Host-2**, **Host-4**, **Host-5**, **Host-6**, and **Host-7** remain completely uninvolved in the transmission, keeping their respective physical links clear for simultaneous data exchanges.

How Star Topology Ensures Reliable LAN Communication

- **Fault Isolation and High Availability:** Because every device connects independently to the central switch, a cable damage event or host failure affects only that single isolated link. All other network nodes continue operating normally without interruption.
- **Collision-Free Data Transfer:** Modern switches operate in full-duplex mode using micro-segmentation. This eliminates data collisions on individual cable runs, providing maximum achievable bandwidth to each host.
- **Scalability and Simplified Maintenance:** Adding, removing, or troubleshooting an individual host requires modifying only a single cabling port at the central switch, preventing network downtime for the remaining devices.

2. Write a Python function named `validate_segment(length_m)` to verify whether a given **UTP cable segment** satisfies the IEEE 802.3 Ethernet standard, which specifies a maximum cable length of **100 meters**. The function should return **True** for valid cable lengths and return **False** along with an appropriate error message for invalid lengths exceeding the permissible limit. Demonstrate the correctness of the program by executing it with at least **three different test cases** and interpret the results.

CODE:

```
def validate_segment(length_m):
    MAX_LENGTH = 100

    if length_m <= MAX_LENGTH:
        return True, "Valid: Cable length is within the 100 m limit."
    else:
        return False, "Invalid: Cable length exceeds the maximum limit of 100 m."

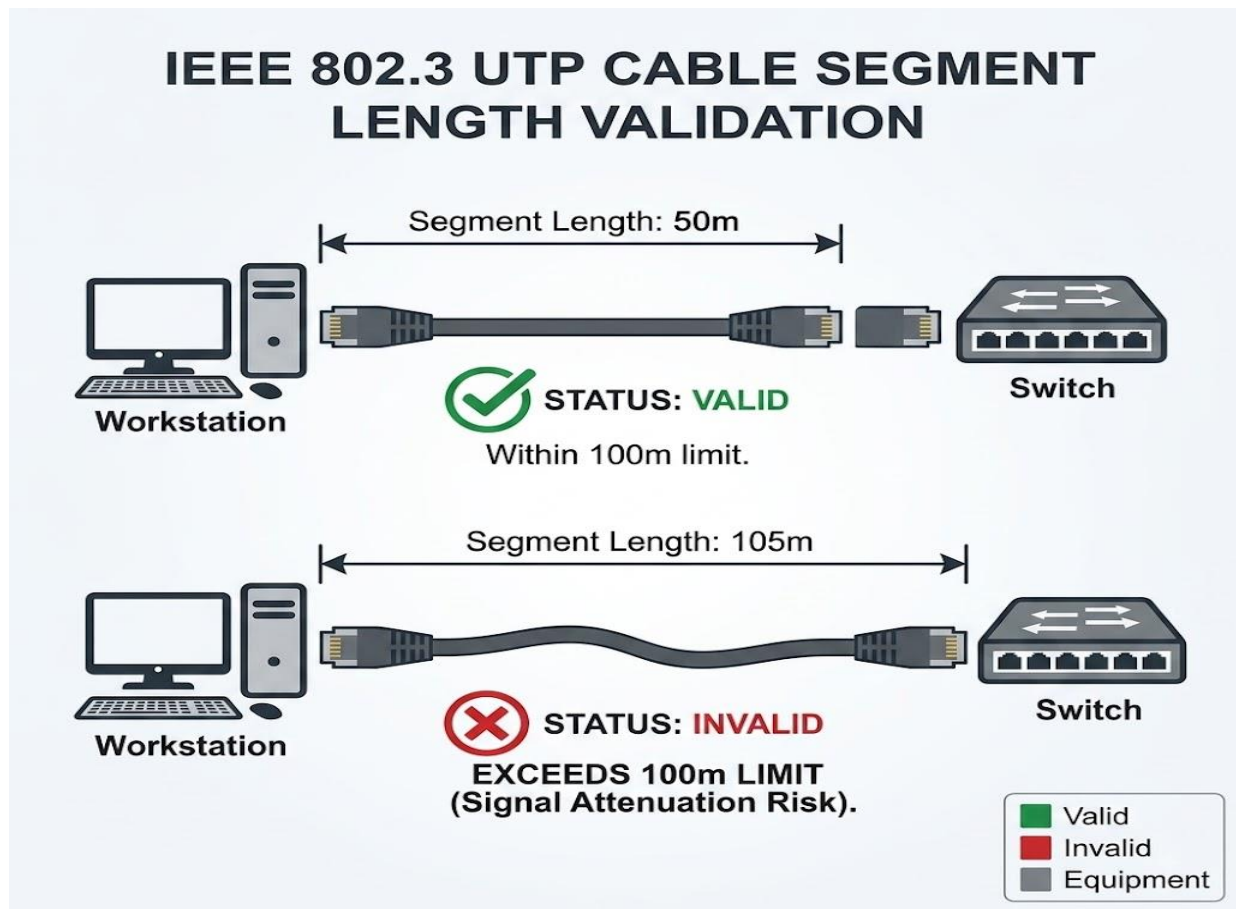
# Test Cases
test_cases = [50, 100, 120]

for length in test_cases:
    result, message = validate_segment(length)
    print("Cable Length:", length, "m")
    print("Result:", result)
    print("Message:", message)
    print("-----")
```

OUTPUT:

```
Output
Cable Length: 50 m
Result: True
Message: Valid: Cable length is within the 100 m limit.
-----
Cable Length: 100 m
Result: True
Message: Valid: Cable length is within the 100 m limit.
-----
Cable Length: 120 m
Result: False
Message: Invalid: Cable length exceeds the maximum limit of 100 m.
-----
```

DIAGRAM:



EXPLANATION:

The diagram visually models the physical layer (Layer 1 of the OSI model) operational outcomes when evaluating copper twisted-pair cable runs against the **IEEE 802.3 Ethernet Standard**. Physical layer transmission limits exist due to electro-physical constraints inherent in transmitting high-frequency electromagnetic signals over solid copper media.

Detailed Breakdown of the Visual Outcomes

Top Scenario: Valid Connection (50m Cable)

- **Signal Attenuation & Signal-to-Noise Ratio (SNR):** In the 50-meter deployment, the electrical signal transmitted from the workstation's Network Interface Card (NIC) reaches the switch port well within acceptable voltage thresholds. Signal attenuation—the reduction in signal strength as energy dissipates into the copper conductor—is minimal.
- **Bit Error Rate (BER):** Because the signal-to-noise ratio remains high, the receiving hardware can easily distinguish binary logic states (1s and 0s). This results in a near-zero Bit Error Rate, eliminating frame corruption caused by signal degradation.
- **Logical Validation:** The function executes the evaluation $50 \leq 100$, evaluating to True. The green checkmark symbolizes optimal data link integrity, full-duplex gigabit throughput, and compliance with ANSI/TIA-568 cabling standards.

Bottom Scenario: Invalid Connection (105m Cable)

- **Excessive Attenuation & Distortion:** The wavy distortion along the 105-meter cable highlights severe signal degradation. As length exceeds 100 meters, high-frequency voltage waveforms experience progressive loss of amplitude and edge sharpness (dispersion), causing the original digital pulses to flatten and blur.
- **Propagation Delay & Timing Violations:** At 105 meters, the round-trip propagation delay increases beyond the maximum time slot allowed for link synchronization and collision domain limits in standard Ethernet framing. This creates timing mismatches, late collisions in half-duplex setups, or link flapping (repeated connection drops) in full-duplex auto-negotiation.
- **Logical Validation:** The function evaluates $105 \leq 100$, evaluating to False. The red 'X' flags a physical layer fault, indicating that packet retransmissions, dropped frames, or total link loss will occur unless a active repeater, switch, or media converter (e.g., fiber optics) is inserted.

Technical Subdivisions of the 100-Meter Channel Limit

The 100-meter total channel distance enforced by IEEE 802.3 and ANSI/TIA-568 standards is architected as follows:

Channel Segment	Maximum Length	Cable Construction Type	Primary Installation Function
Work Area Patch Cord	5 meters	Stranded Copper	Connects end-user workstation/device to the wall outlet.
Permanent Link	90 meters	Solid Copper Core	In-wall/ceiling conduit run from wall outlet to telecommunications room.
Closet Patch Cord	5 meters	Stranded Copper	Connects patch panel in equipment rack to the network switch port.
Total Channel	100 meters	Mixed Construction	Complete end-to-end connection satisfying IEEE 802.3.

- **90m Permanent Link:** Composed of solid-core copper wires run inside walls, ceiling trays, and conduits. Solid copper provides lower resistance over long spans.
- **10m Total Patch Cables:** Combined length of stranded-core patch cords (usually 5 meters at the user desk and 5 meters in the server rack). Stranded cables offer physical flexibility for frequent bending but exhibit up to 20% higher signal attenuation per meter than solid-core wire.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Understanding	15%	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
Algorithm Design	20%	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm
Code Quality	15%	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
Functionality	20%	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
Error Handling	10%	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
Efficiency	10%	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
Documentation & Comments	5%	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
Testing & Debugging	5%	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done